

| SystemC Manual Clock Gating within DUT

Refer to Matchlib example 59_idle_signal

<https://github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG>

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17 Apr 2026

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SystemC Manual Clock Gating within DUT – 59_idle_signal

- Simple example manual clock gating within DUT and automatic generation of idle signal in the RTL.
- This is new in 2026.1
- Methodology goals:
 - Locate clock gating cells (CGIC cells) at appropriate places within DUT hierarchy.
 - Accurately model effects of idle signals, clock gating in both pre-HLS and post-HLS sims.
 - Hard requirement: Functionality of DUT must be identical in cases where clock gating is or is not used.
- For pre-HLS idle signal, cannot use automatic idle signal generation, can use explicit modeling.
 - In this example we show explicit modeling of the idle signal in the pre-HLS sim, but this is not required.
- For large subsystems, likely would have many local clock gating modules and idle signals.

DUT Top Level

- On line 105 we instantiate the clock gate module.
- On line 119 the sub block generates the idle signal.
- On line 110 we feed the idle signal into the clock gate module.
- On line 114 we feed the gated_clk into the sub block.

```
93 #pragma hls_design top
94 class dut : public sc_module
95 {
96 public:
97     sc_in<bool> SC_NAMED(clk);
98     sc_in<bool> SC_NAMED(rst_bar);
99
100     Connections::Out<uint32> SC_NAMED(out1);
101     Connections::In <uint32> SC_NAMED(in1);
102     Connections::In <uint32> SC_NAMED(in2);
103     SC_SIG(bool, idle);
104     SC_SIG(bool, gated_clk);
105     clock_gate_module SC_NAMED(clock_gate_module1);
106
107     sub SC_NAMED(sub1);
108
109     SC_CTOR(dut) {
110         clock_gate_module1.idle_in(idle);
111         clock_gate_module1.clk_in(clk);
112         clock_gate_module1.clk_out(gated_clk);
113
114         sub1.clk(gated_clk);
115         sub1.rst_bar(rst_bar);
116         sub1.out1(out1);
117         sub1.in1(in1);
118         sub1.in2(in2);
119         sub1.idle(idle);
120     }
121 };
```

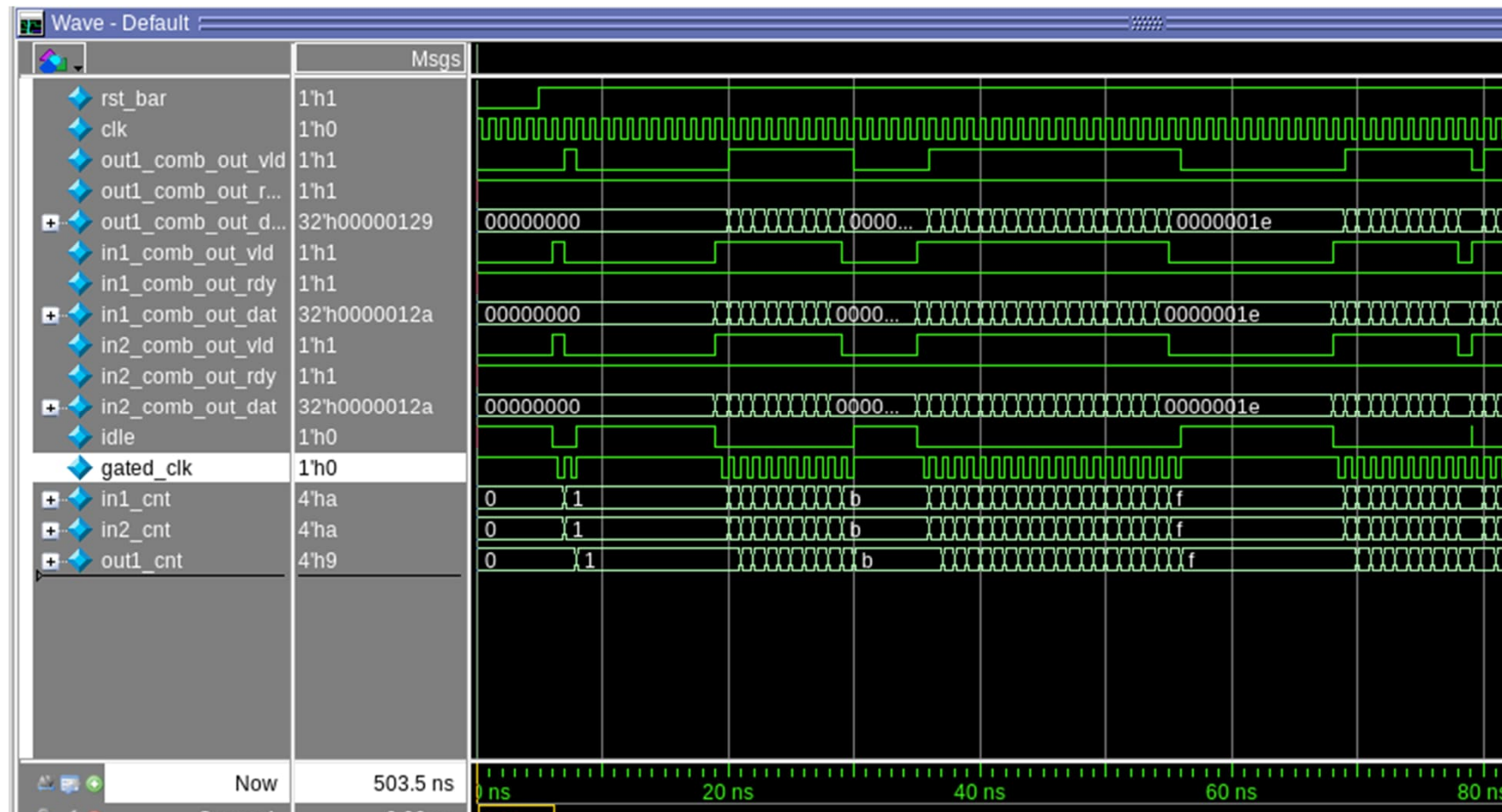
Clock Gate Module

- On line 28 we use the `ac_blackbox` feature to instantiate the Verilog CGIC cell.
- On line 26 we have a `SC_THREAD` for the clock gate module for pre-HLS sim.

```
10 class clock_gate_module : public sc_module {
11 public:
12     sc_in<bool> SC_NAMED(idle_in);
13     sc_in<bool> SC_NAMED(clk_in);
14     sc_out<bool> SC_NAMED(clk_out);
15
16     // This model does work in all scenarios (pre-HLS, post-HLS sims in Questa, OSCI, etc).
17     // It is not the most obvious implementation..
18
19     // This is a SystemC simulation only model. In the RTL, this module is replaced
20     // with a CGIC cell (clock gate insertion cell) using the ac_blackbox construct.
21     // The key requirement of this model is to introduce no delta delays into the gated clock
22     // as compared to the input clock.
23
24     SC_CTOR(clock_gate_module) {
25 #ifndef __SYNTHESIS__
26         SC_THREAD(main);
27 #else
28         ac_blackbox()
29         // netlist info
30         .entity("clock_gate_module")
31         .architecture("bb")
32         .verilog_files("latch_based_clock_gating_comb.v")
33         .vhdl_files("latch_based_clock_gating_comb.vhd")
34         .outputs("clk_out")
35         .clock_name("clk_in")
36         .clock_gated(true)
37         // area/timing
38         .area(12.3456)
39         .delay(0.0)
40         .latency(1)
41         .timing_units("ns")
42         .end();
43 #endif
44     }
```

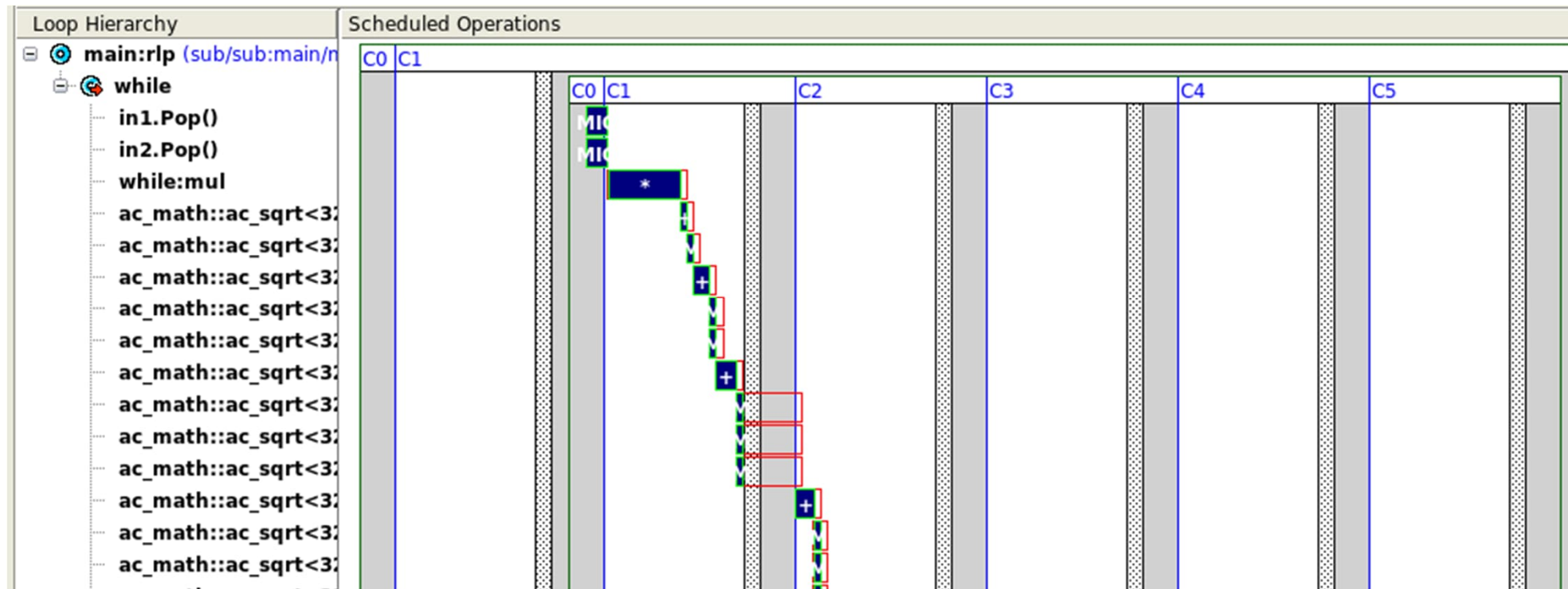
Pre-HLS Simulation

- We disable the `#define AUTOMATIC_IDLE` so the idle signal and gated_clk are active.
- We see when idle is high, the gated_clk is pulled high.



Post-HLS RTL DUT Pipeline

- Post-HLS RTL DUT has 5 stage pipeline
- Pipeline must be flushed before DUT can go idle
- This alters the clock gating timing between pre-hls and post-hls models.



Post-HLS Simulation

- We enable the `#define AUTOMATIC_IDLE` so the `gated_clk` is active via Catapult's idle signal
- We see when idle is high, the `gated_clk` is pulled high.

